

Pierre Maillot

PhD and Engineer in Knowledge Graphs and AI

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Current Position

- 2024 – Present **Research Engineer**, *France 2030: P16*, INRIA Sophia Antipolis, (24 months)
Pre-industrialization of the open-source knowledge graph engine Corese, in collaboration with :Probabl.
Co-supervision of an internship focused on developing a GraphRAG application based on RDF knowledge graphs.

Experience

- 2021 – 2024 **Postdoctoral Researcher**, *ANR Project DeKaLoG*, INRIA Sophia Antipolis, (42 months)
Design and implementation of IndeGx, an open-source generator of indexes for knowledge graphs and their metadata.
Collaboration with LIRIS on the design and implementation of a method for evaluating the transparency and FAIRness of knowledge graphs.
- 2017 – 2019 **Technology Transfer Engineer**, *Projet SEMELEV*, SATT Ouest Valorisation, (18 months)
Industrialization of the knowledge graph exploration prototype SPARKLIS toward the ASKELYS startup.
- 2015 – 2017 **Postdoctoral Researcher**, *Projet ANR IDFraud*, IRISA/Uni. Rennes 1, équipe SemLIS, Rennes, (24 months)
Conception and implementation of FORMULIS, a data generation system for knowledge graphs. Experimentation with the French Gendarmerie.
Design of a data-mining-based method for evaluating the evolution of knowledge graphs.
- 2012 – 2015 **PhD in Computer Science, University of Angers**, *26 Novembre 2015, UFR Sciences d'Angers*, Funded by the FUI TIMCO project
- Title* **Nouvelles méthodes pour l'évaluation, l'évolution et l'interrogation des bases du Web des données**
New methods for the evaluation, evolution and querying of web of data knowledge graphs
- Committee:* Chair: Francis Rousseaux, Professor, Université de Reims
Reviewers: Juliette Dibie, Prof., AgroParisTech/INRA; Mohand-Saïd Hacid, Prof., LIRIS
- Supervision* PhD Advisor: Stéphane Loiseau, Professor, LERIA. Co-supervisors: David Genest, Associate Professor, LERIA; Thomas Raimbault, Associate Professor, ESILV.

Skills

General	Autonomy, collaborative work, communication with domain experts, public outreach, adaptability, open-mindedness, rigor, mobility
Programming	Java, Python, Typescript, JavaScript, NodeJS, C/C++, Bash, Git, Docker
Others	Knowledge graphs, Data Mining, Pattern mining, FAIR, Open Data, Design Patterns, SOLID principles
Language	English (fluent), French (native)

Projects

Github	https://github.com/MaillPierre Corese, open source knowledge graph engine (Java) [2] IndeGx, online endpoint indexation system for metadata (Java/Typescript). License GPL-3.0. [6] Metadatamatic, Metadata generation website for knowledge graphs (Typescript). Apache-2.0. [8] KartoGraphI, website showcase of the state of the art of the publicly accessible endpoint indexed by IndeGx (Javascript). [10, 11] FORMULIS, dynamic guided knowledge graph data creation system (Java/Javascript). CECILL V2.1/GPL-3.0. [9] Data mining technique to evaluate the similarity of knowledge graphs (Java). [3] Update evaluation and version comparison for knowledge graphs based on similarity (Java).
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Expertises

Knowledge graphs	Metadata extraction for knowledge graph annotation and query optimisation [2, 4, 6, 8].
Data mining	Structural similarity evaluation of knowledge graphs at scale [1, 9].
Data quality	Knowledge graph update evaluation based on similarity[3]. Quality and transparency evaluation of knowledge graphs [2, 4, 5].

Public service

Reviews	FCA4AI 2016 (Workshop ECAI), EGC 2017, ECML/PKDD (2017, 2022), SAC18 (SWA track), eKNOW (2021, 2022, 2023), MEPDaW (2021, 2023), Journal of Web Engineering 2022, ISWC 2023, Artificial Intelligence Review 2023, ESWC 2024, ECAI 2025
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Publications

International journals with review committee

- [1] Carlos Bobed, Jorge Bernad, and Pierre Maillot. Language-Model Based Informed Partition of Databases to Speed Up Pattern Mining.
In *SIGMOD* 2024. CORE A++, GGS 1/A++. 27 pages.
- [2] Pierre Maillot, Olivier Corby, Catherine Faron, Fabien Gandon, Franck Michel. IndeGx: A model and a framework for indexing RDF knowledge graphs with SPARQL-based test suits.
In *Journal of Web Semantics* 2023. 20 pages.
- [3] Carlos Bobed, Pierre Maillot, Peggy Cellier, Sébastien Ferré. Data-driven assessment of structural evolution of RDF graphs.
In *Semantic Web Journal* Volume 11, Number 5, 2020. 22 pages.

International conferences with review committee

- [4] Pierre Maillot, Jennie Andersen, Sylvie Cazalens, Catherine Faron, Fabien Gandon, Philippe Lamarre, Franck Michel An Open Platform for Quality Measures in a Linked Data Index.
In *WWW (Companion Volume)*, resource track, 2024. CORE A++, GGS 1/A++.
- [5] Jennie Andersen, Sylvie Cazalens, Philippe Lamarre, Pierre Maillot A Framework to Assess Knowledge Graphs Accountability. CORE B, GGS 3/B.
- [6] Pierre Maillot, Olivier Corby, Catherine Faron, Fabien Gandon, Franck Michel Metadatamatic: A Web application to Create a Dataset Description.
In *WWW (Companion Volume)*, poster, 2023. CORE A++, GGS 1/A++.
- [7] Aline Menin, Pierre Maillot, Catherine Faron, Olivier Corby, Carla Maria Dal Sasso Freitas, Fabien Gandon, Marco Winckler LDViz: A Tool to Assist the Multidimensional Exploration of SPARQL Endpoints.
In *WEBIST (Revised Selected Papers)*, 2023. CORE C.
- [8] Pierre Maillot, Olivier Corby, Catherine Faron, Fabien Gandon, Franck Michel KartoGraphI: Drawing a Map of Linked Data.
In *Extended Semantic Web Conference*. Springer, 2022. CORE A, GGS 2/A-. Demo, 4 pages
- [9] Pierre Maillot, Carlos Bobed. Measuring structural similarity in the Semantic Web.
In *ACM Symposium on Applied Computing (ACM SAC)*, Pau, France, 2018. CORE B, GGS 2/A-, taux d'acceptation: 25%. Full paper, 8 pages.
- [10] Pierre Maillot, Sébastien Ferré, Peggy Cellier, Mireille Ducassé and Franck Partouche. Nested Forms with Dynamic Suggestions for Quality RDF Authoring.
In *International Conference on Database and Expert Systems Applications (DEXA)*, Lyon, France, 2017. CORE B, GGS 3/B, taux d'acceptation: 24%. Short paper, 10 pages, Springer.
- [11] Pierre Maillot, Sébastien Ferré, Peggy Cellier, Mireille Ducassé and Franck Partouche. FORMULIS: Dynamic Form-Based Interface For Guided Knowledge Graph Authoring.
In *International Conference on Knowledge Engineering and Knowledge Management (EKAW)*, Bologne, Italie, 2016. CORE B, GGS 3/B, taux d'acceptation: 30%. Poster/Demo, 4 pages, Springer.

CORE evaluation: <http://portal.core.edu.au/conf-ranks/>.

GGs evaluation: <http://valutazione.unibas.it/gii-grin-scie-rating/>.

Complete list of publications on my personal webpage: <https://w3id.org/people/pierremaillot>